02-719 Week 8

Genomics and Epigenetics of the Brain

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Recap: Convergent Evolution

- The same phenotype evolves **independently** in separate lineages
- Each origin is a natural replicate of the same experiment
  - If a gene matters for the trait, it may respond the same way each time
- The question: Does a gene's **rate of evolution** shift on the branches where are trait evolved?

Species are not independent data points

Naive approach:

- Correlate a gene's sequence with the trait across species
- Closely related species share both, so one origin can look like many
- $\rightarrow$ inflated significance

Phylogenetic approach:

- Work on branches of the tree, not species
- Count each independent origin once
- $\rightarrow$ signal must repeat across lineages

RER Coverage

RER = Relative evolutionary rates. **RERcoverage** is an R workflow that does the following analysis:

- Inputs: gene trees with a fixed topology
- Relative evolutionary rates (RERs)
- Encoding the phenotype (binary vs continuous traits)
- Testing for association
- From gene lists to biology

RERcoverage Workflow

1. Alignments: one per gene
2. Gene trees: fixed topology, gene-specific branch lengths
3. RERs: relative rate on every branch
4. Phenotype paths: trait mapped onto branches
5. Correlation: RER vs. trait, per gene
6. Enrichment: Pathways / gene ontology

Each gene gets one test. The output is a ranked list of genes whose rates shift with the trait across the tree.

Step 1: Inputs, one tree per gene

- Start from a multiple alignment of orthologs for each gene
- Fix the **species tree topology**; estimate only the branch lengths
  - Branch length = substitutions per site on that branch
  - Species missing the gene are pruned for that gene
- readTrees() combines all genes
  - Builds a **master tree**: average branch lengths across genes

[insert image]

Why not compare raw branch lengths?

$$\text{branch length} = \text{time} \times \text{lineage rate} \times \text{gene-specific rate}$$

- Time and lineage rate are shared for all genes
- We only want the gene-specific rate
- Therefore, comparing raw branch lengths might not give us what we are looking for.

Step 2: Fit the Expected rate

- Graph branches for the gene of interest on the y-axis, and master tree branch length on the x-axis.
- Run regression on the data. The line is the **expected** length for the given gene. Points above the line are faster than expected, while points below the line are slower than expected.
- The residuals (distance from point to line) are the RERs. (RER = observed - expected)
- RERconverge asks whether the residuals are statistically significant?

Making RERs Robust

- Problem: long branches are noisier (and will dominate a plain regression)
- Fixes:
  - Square root transform the branch lengths
  - Weighted regression
  - Scale RERs across genes

Step 3: Encoding a Binary Trait

- Foreground = branches where the trait is present
- Background = all other pranches
- Input: a list of foreground species

Step 3: Encoding a Continuous Trait

- Every species has a value (e.g., maximum lifespan)
- **Ancestral state reconstruction** estimates values at internal nodes
- By default, each branch gets the **change** in trait along it.
- Correlate trait change wiht RER across branches.

Step 4: Testing each gene for association

- For binary: `correlateWithBinaryPhenotype()`. Uses Kendall's Tau correlation between RER, 0/1 by default.
- For continuous: `correlateWithContinuousPhenotype()`. Uses Pearson correlation by default.
- Genes need enough species and foreground branches.

Rho (ρ) represents the correlation coefficient between the RER of a genetic region and a specific phenotypic trait.

- $\rho > 0$, gene is faster in foreground. (function may be no longer needed, or positive selection. Example: eye genes in blind mammals)
- $\rho < 0$, gene is slower in foreground. (increased constraint: the function matters more.)

Step 5: Do we Trust p-values?

1. Simulate: a null trait on the same tree
2. Match: same number and structure of foreground species
3. Re-test: every gene against the null trait
4. Repeat: build an empirical p-value per gene.

- Parametric p-values assume branches are independent; they aren't
- **Permutations** = permutation + simulation that keeps the phylogeny
- Expensive: many reruns of the whole analysis.

p-values can be off in either direction: too many hits or too few hits.

Step 6: From Gene lists to Biology

- Rank-based (whole list)
  - Score every gene: $\text{sign}(\rho) \times -\log_{10}(p)$
  - Do gene sets sit at the top or bottom of the ranking?
- Over-representation (top hits)
  - Take genes passing a cutoff, split by sign
  - Test GO terms against a background

Note that background includes all genes that were tested, not the whole genome.